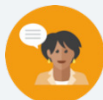


## 1.7 Evaluating Exponential Expressions

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                      |                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.                                                                                          |                                                                                                                                      |  Learning Targets                                                                                                                                                                                | <ul style="list-style-type: none"><li>I can evaluate numerical expressions involving rational exponents.</li><li>I can generate equivalent numerical expressions involving rational exponents.</li></ul> |
|  Benchmark Coherence | Building On<br>MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.                          |                                                                                                                                      | Working Toward<br>N/A                                                                                                                                                                                                                                                               |                                                                                                                                                                                                          |
|  Essential Questions | <ul style="list-style-type: none"><li>How can I use the Laws of Exponents to help me evaluate numerical expressions?</li><li>How can the relationship between rational exponent form and radical expression form help me evaluate numeric expressions without a calculator?</li></ul>                                                   |                                                                                                                                      |                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                          |
|  Lesson Narrative    | In this lesson, students will extend their use of the Laws of Exponents to evaluate numerical expressions that are presented in rational exponent and/or radical expression form. Students will work together to practice how to evaluate these expressions and be able to do them on their own with accuracy by the end of the lesson. |                                                                                                                                      |                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                          |
|  Component Summary | 1.7.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                 | Perfect squares puzzle activity ( <i>SE p. 37</i> )                                                                                  | 1.7.3 Activity (20 min.)                                                                                                                                                                                                                                                            | Equivalent expressions with exponents maze activity ( <i>SE p. 38</i> )                                                                                                                                  |
|                                                                                                       | 1.7.2 Collaboration (10 min.)                                                                                                                                                                                                                                                                                                           | Exploring the simplification of expressions with exponents ( <i>SE p. 37</i> )                                                       | 1.7.4 Your Turn! (10 min.)                                                                                                                                                                                                                                                          | Evaluating expressions in rational exponent form without a calculator ( <i>SE p. 39</i> )                                                                                                                |
|                                                                                                       | 1.7.5 Wrap-Up (~5 min.)                                                                                                                                                                                                                                                                                                                 | Self-reflection activity about the things they are unsure about from the lesson and what they will do to improve ( <i>SE p. 39</i> ) |                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                          |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                |                                                                                                                                      | Materials                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                          |
|                                                                                                       | Key Terms: N/A<br><br>Additional Terms: <ul style="list-style-type: none"><li>Rational Exponent</li><li>Radical Form</li><li>Equivalent Expression</li></ul>                                                                                                                                                                            |                                                                                                                                      | <ul style="list-style-type: none"><li>N/A</li></ul>                                                                                                                                                                                                                                 |                                                                                                                                                                                                          |
|                                                                                                       |                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                      | Additional Preparation <ul style="list-style-type: none"><li>Work through the maze before you have students do it. When completing it, time yourself so you can determine an appropriate amount of time your students will need to complete it or get as far as they can.</li></ul> |                                                                                                                                                                                                          |

| <br>Teacher Tips               | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Things to Consider                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                 | <ul style="list-style-type: none"><li>Students may not remember to convert between forms where appropriate.</li><li>Students may not make the connection between the two forms.</li><li>Students may put the numerator of the fraction in the exponent as the index of the radical form.</li><li>Students may make calculation errors when determining the value of the expression.</li></ul>                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"><li>This lesson requires students to do evaluation without a calculator and the use of perfect squares table. The table that will be created in question 1 of the 1.7.1 Warm-Up can be a valuable reference tool for visual learners in the future.</li><li>Consider having this table created for students to physically fill out and keep somewhere in their math tools folder or file for future use. As an alternative, consider creating the table on an Anchor Chart somewhere for students to reference when needed.</li><li>Consider discussing common errors from 1.6.6 Wrap-Up before 1.7.2 Collaboration.</li><li>Consider not allowing a calculator on the lesson.</li></ul> |
|                                                                                                                 | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                                                                                 | <b>Which One Doesn't Belong?</b><br>1.7.3 Activity provides students with multiple scenarios where students must decide which one of the adjacent spaces contains an expression that is equivalent to the current space. Students can engage in this protocol to determine which expression does not belong, using the conditions of equivalency.                                                                                                                                                                                                                                                                                                                                                                | <b>MA.K12.MTR.1.1: Participate in Effortful Learning</b><br>1.7.3 Activity provides students with a challenging maze where students must apply all they have learned about equivalent forms with exponents to determine the next step to take in the equivalency maze. To do this, students will need to modify methods and choose the best tools needed to persevere while solving this task.                                                                                                                                                                                                                                                                                                                             |
| <br>Differentiate Instruction | Consider placing students in groups and assigning groups to each number of the 1.7.2 Collaboration. By differentiating which groups engage with each portion of the activity, the lesson can become a jigsaw in which students are responsible for sharing their learning with others.<br>For 1.7.1 Warm-Up, consider having the table from question 1 created for students to physically fill out and keep somewhere in their math tools folder or file for future use.<br>1.7.4 Your Turn! provides students an opportunity to show what they know related to the content of this lesson. How students perform on this activity can provide differentiation data for students who may need additional support. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Lesson Notes:                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## 1.7.1 Warm-Up: Perfect Squares Puzzle

**Component Overview:** Students will complete a puzzle based on perfect squares.



### 1.7.1 Warm-Up: Perfect Squares Puzzle

- Complete the table by finding the square of each whole number.

| Whole Number | Square     | Whole Number | Square       | Whole Number | Square       |
|--------------|------------|--------------|--------------|--------------|--------------|
| 1            | $1^2 = 1$  | 8            | $8^2 = 64$   | 15           | $15^2 = 225$ |
| 2            | $2^2 = 4$  | 9            | $9^2 = 81$   | 16           | $16^2 = 256$ |
| 3            | $3^2 = 9$  | 10           | $10^2 = 100$ | 17           | $17^2 = 289$ |
| 4            | $4^2 = 16$ | 11           | $11^2 = 121$ | 18           | $18^2 = 324$ |
| 5            | $5^2 = 25$ | 12           | $12^2 = 144$ | 19           | $19^2 = 361$ |
| 6            | $6^2 = 36$ | 13           | $13^2 = 169$ | 20           | $20^2 = 400$ |
| 7            | $7^2 = 49$ | 14           | $14^2 = 196$ | 21           | $21^2 = 441$ |

- Place a digit in each blue box so that the principal square root is a whole number, and the expression is as close to 0 as possible. Use only the digits 1 to 9, at most one time each.  
Reminder: The principal square root is the positive square root.

$$\sqrt{\boxed{2} \boxed{5} \boxed{6}} - \sqrt{\boxed{8} \boxed{1}} - \sqrt{\boxed{9}}$$



The table that will be created in question 1 can be a valuable reference tool for visual learners in the future. Consider having this table created for students to physically fill out and keep somewhere in their math tools folder or file for future use. As an alternative, consider creating the table on an Anchor Chart somewhere for students to reference when needed.



Consider allowing students to work together on question 2 in this activity. Students may need a reminder of what the directions mean, as this is the second time, they have encountered this type of problem in the material so far.

## 1.7.2 Collaboration: Simplifying Expressions With Exponents

**Component Overview:** Students will work with a partner or small group to explore simplifying expressions with exponents. For this activity, release students to work with their partner for all five questions, prior to bringing the class back together to review the answers.

### 1.7.2 Collaboration: Simplifying Expressions with Exponents

1. With a partner, write the expression  $(27)^{\frac{2}{3}}$  in three equivalent forms.

|                      |                        |                    |                  |
|----------------------|------------------------|--------------------|------------------|
| $(27)^{\frac{2}{3}}$ | $(27^{\frac{1}{3}})^2$ | $(\sqrt[3]{27})^2$ | $\sqrt[3]{27^2}$ |
|----------------------|------------------------|--------------------|------------------|

2. Explain which equivalent form makes it easiest to find the value of  $(27)^{\frac{2}{3}}$  without a calculator.

*$(\sqrt[3]{27})^2$  allows me to determine the cube root of 27, which is 3 and then square that value which would be 9. Each of those calculations can be done using mental math.*

3. Find the value of  $(27)^{\frac{2}{3}}$  without a calculator.  
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4. Verify the value of  $(27)^{\frac{2}{3}}$  using your calculator.  
*Students should be able to show the answer on their calculator is the same as their answer above.*

5. For each expression, write an equivalent radical expression. Then, evaluate the expression using your calculator.

| Expression                       | Equivalent Radical Expression  | Value of the Expression |
|----------------------------------|--------------------------------|-------------------------|
| $(144)^{-\frac{3}{2}}$           | $\frac{1}{\sqrt{144^3}}$       | $\frac{1}{1,728}$       |
| $-(-64)^{\frac{1}{3}}$           | $-\sqrt[3]{-64}$               | 4                       |
| $(225)^{-\frac{1}{2}}$           | $\frac{1}{\sqrt{225}}$         | $\frac{1}{15}$          |
| $-9^{\frac{5}{2}}$               | $-\sqrt{9^5}$                  | -243                    |
| $(\frac{81}{256})^{\frac{3}{4}}$ | $\sqrt[4]{(\frac{81}{256})^3}$ | $\frac{27}{64}$         |



Students' fluency on equivalent forms should be increasing. As you circulate, look for students who still have to refer to previous lessons to know how to write equivalent expressions. Make note of these students and consider utilizing the activity to support these students using small group instruction. Consider asking these questions to help establish student understanding:



- How did you go about deciding which three equivalent forms you were going to use?
- Why did you choose the form you did for question 2?



Students may write different equivalent radical expressions. For some, determining the value from "their" equivalent expression may be difficult without a calculator. Instead of giving the student a calculator, ask the student to find another way to write their equivalent radical expression so that finding the value would be easier. For example,  $(144)^{-\frac{3}{2}}$  written as  $\frac{1}{\sqrt{144^3}}$  is harder than  $\frac{1}{(\sqrt{144})^3}$ . This helps the student understand why it is valuable to know how to rewrite numeric expressions with rational expressions using a variety of equivalent expressions.

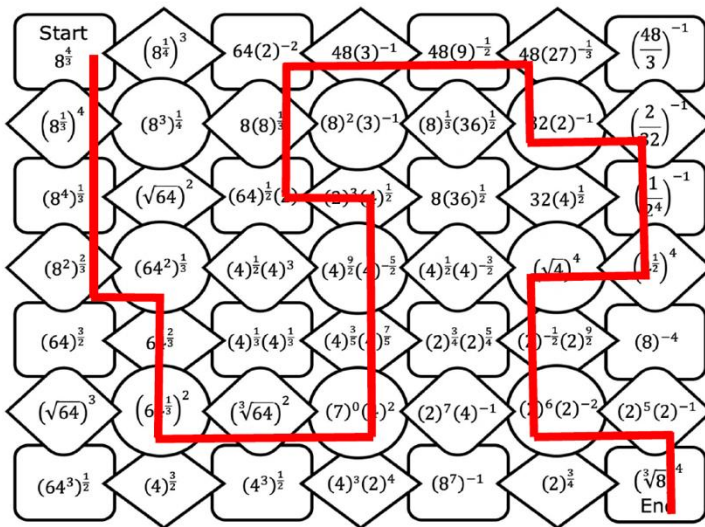
### 1.7.3 Activity: Exponents Maze

**Component Overview:** Students will complete a maze where they go from start to finish by only connecting to an adjacent equivalent exponent. For this activity, review the narrative information from the question prior to releasing students to do questions 1 and 2.



### 1.7.3 Activity: Exponents Maze

1. Work with a partner to complete the maze. Beginning at the "start" box, connect each expression to an equivalent expression to form one continuous path to the "end" of the maze. Move only horizontally or vertically.



2. What is the value of the expression in each box that created the path through the maze?

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*When reviewing the instructions and context for question 1, consider providing students with strategies to leverage when working together as partners. For example, each partner takes one of the adjacent expressions and evaluates them. Each partner shares what they evaluated to see if either of them gets the same answer as the prior expression. This will help students save time when trying to complete this activity.*



*For a challenge, allow students who are doing well with the content to work on this maze independently if time allows.*



*Consider using a timer during this activity. If left unstructured, this activity could take the remainder of the class period. Consider applying a time limit to this activity. The purpose is to provide practice of identifying equivalent expressions, not strictly completing the maze.*

## 1.7.4 Your Turn!

**Component Overview:** Students will work independently to practice evaluating expressions in rational exponent form without a calculator. For this activity, release students to complete the problems in the table on their own. Bring students back together to review the answers to the activity.



### 1.7.4 Your Turn!

1. Evaluate each expression without a calculator.

|                                                               |                                                                                     |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------|
| $(100)^{\frac{1}{2}} \cdot (2^3)$<br>80                       | $(8^{-1})\left(4^{\frac{5}{2}}\right)$<br>4                                         |
| $(81^6)^{\frac{1}{24}}$<br>3                                  | $(125)^{\frac{2}{3}}$<br>25                                                         |
| $(49^{\frac{1}{2}})(3^{-2})$<br>$\frac{7}{9}$                 | $-(3^{12})^{\frac{1}{4}}$<br>-27                                                    |
| $(2^4)\left(8^{-\frac{2}{3}}\right)$<br>4                     | $\left(\frac{15}{45}\right)^{-4}$<br>81                                             |
| $-\left(\frac{3}{48}\right)^{\frac{3}{2}}$<br>$-\frac{1}{64}$ | $\left(\frac{144}{169}\right)^{-\frac{1}{2}}$<br>$1\frac{1}{12}$ or $\frac{13}{12}$ |



Consider using this time to work with a small group of students who may need additional support based on their performance throughout this lesson and prior lessons in this unit. In the small group, use squares 1 and 3 and allow students to work on squares 2 and 4 on their own and show you their answers and work so you can guide them through the process when needed.



Consider allowing students to check their work with a partner if they finish before the allotted time. During the share-out of the answers, have students explain how they got their answers and any methods or strategies that they leveraged.



- Which problem did you find the most challenging to evaluate?
- Were there any that, after seeing the correct answer, you are still unclear as to how that answer was correct?

## 1.7.5 Wrap-Up: Reflection

**Component Overview:** Students will complete a self-reflection activity about the things they are unsure about from the lesson and what they will do to improve. This activity should be done independently to get an accurate assessment of where students feel unsure related to the content of this lesson.



### 1.7.5 Wrap-Up: Reflection

Complete the following statements about today's lesson.

*Answers will vary.*

1. Today I am still unsure about .....
  
2. To fill in this gap I intend to .....



*Since this Wrap-Up activity is about a student's self-reflection, this data can be coupled with their overall academic performance on this lesson to create lists of students, by combination profile, who may need additional differentiated support.*